

Joshua Fan

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SUMMARY

PhD candidate (Cornell University, Computer Science) with 7 yrs experience leading applied ML projects – from data engineering, understanding customer needs, research, model design, training, debugging, to evaluation. Open to ML Scientist or Engineer roles where I can use my AI/ML expertise for real-world impact. Available to start in **Feb 2026**.

EDUCATION

PhD Computer Science, Cornell University – *Ithaca, NY* **Aug 2019 – Dec 2025 (expected)**

- Thesis: Spatio-Temporal Deep Learning with Limited Labels: Applications in Computational Sustainability
- Advisor: Prof. Carla Gomes GPA: 4.10/4.3

MS Computer Science, University of Washington – *Seattle, WA* **Mar 2017 – June 2019**

- Advisor: Prof. Sreeram Kannan GPA: 3.84/4.0

BS Computer Science, University of Washington – *Seattle, WA* **Sep 2013 – Mar 2017**

- Advisor: Prof. Richard Anderson GPA: 3.97/4.0 (summa cum laude)

RESEARCH EXPERIENCE

Research Assistant, Cornell University (Computational Sustainability) – *Ithaca, NY* **Aug 2019 – Present**

- Integrated **scientific models** into **deep learning** in an **interpretable** way to simulate the soil carbon cycle. Our approaches (SciReN/BINN) respect scientific laws, infer unlabeled biogeochemical processes, and help scientists gain insight into how input variables influence biogeochemical processes [1][10].
- Designed novel **self-supervised (contrastive) learning** methods for **satellite images** to map fish farms in the Amazon. Used masked pooling and custom augmentations to overcome difficult **distribution shifts**, increasing F1 by 20% [8].
- Created an **interpretable spatiotemporal deep learning** framework (GNN-RNN) to forecast crop yields in US counties [5]. With only public weather/soil data, we often achieve better accuracy than USDA's expensive on-the-ground forecasts.
- Developed novel **Transformer** variants which emphasize locality and relative positioning, reducing prediction error (RMSE) by an average of 18% on multiple **time series regression** tasks over the previous state-of-the-art [9].
- Devised a **weakly-supervised U-Net** method, which can predict at extremely **fine resolutions** (30m) even when training labels are only available at a much coarser resolution (3km). We use this to predict SIF (vegetation productivity) from **remote sensing** data at unprecedented fine resolutions, reducing RMSE by 9% over the strongest baselines [4].

Research Assistant, University of Washington (Information Theory Lab) – *Seattle, WA* **Mar 2017 – June 2018**

- Developed scalable algorithms inspired by Latent Dirichlet Allocation and matrix factorization to discover cell types and find structure in large single-cell RNA-seq datasets (over 1 million cells) ([Poster](#), [Paper](#), [Code](#))

Research Assistant, University of Washington (Computing for Development Lab) – *Seattle, WA* **Mar 2015 – June 2016**

- Redesigned a survey app which helps health workers collect patient data and suggest follow-up actions

INDUSTRY EXPERIENCE

Research & Development Intern, Kitware (*Computer Vision team*) – *Clifton Park, NY* **June 2024 – Aug 2024**

- Researched techniques to mitigate loss of plasticity in deep continual learning – to allow pretrained models to learn incrementally from new datasets and tasks

NLP Research Intern, Docugami (*AI document engineering*) – *Kirkland, WA* **Mar 2018 – Sept 2018, Jan 2019 – Aug 2019**

- Researched and implemented state-of-the-art NLP algorithms (including topic modeling, clustering, and question-answering techniques), and adapted them in novel ways for enterprise document analysis

Software Engineer Intern, Meta (*Integrity Computer Vision Team*) – Seattle, WA

Sept 2018 – Dec 2018

- Trained a clip-based 3D CNN to detect graphic and violent content in videos, outperforming previous approaches

Software Engineer Intern, Meta (*Search, Whole Page Ranking Team*) – Menlo Park, CA

Sept 2017 – Dec 2017

- Trained sequence neural networks to predict search behavior based on recent query history; improved user click rate

Software Engineer Intern, Meta (*Search Indexing Team*) – Seattle, WA

June 2016 – Sept 2016

- Built a web tool to help engineers debug and simulate changes to the search indexing pipeline

Software Design Engineer Intern, BitTitan – Bellevue, WA

June 2015 – Sept 2015

- Built infrastructure to test mailbox migrations in memory; optimized SQL queries (reducing runtime by 60%)

PUBLICATIONS (peer-reviewed conference and journal papers; * denotes equal contribution)

[1] Scientifically-Interpretable Reasoning Network (SciReN): Discovering Hidden Relationships in the Carbon Cycle.

Joshua Fan*, Haodi Xu*, Feng Tao*, Md Nasim, Marc Grimson, Yiqi Luo, Carla P. Gomes.

AAAI-26: *AAAI Conference on Artificial Intelligence (AI for Social Impact track)*, 2026. **(Acceptance rate: 24.1%)**

[2] LabelKAN - Kolmogorov-Arnold Networks for Inter-Label Learning: Avian Community Learning.

Marc Grimson, Joshua Fan, Courtney L. Davis, Dylan Van Bramer, Daniel Fink, Carla P. Gomes.

AAAI-26: *AAAI Conference on Artificial Intelligence (AI for Social Impact track)*, 2026. **(Acceptance rate: 24.1%)**

[3] Towards Sustainable Aquaculture in the Amazon.

Felipe S. Pacheco, Sebastian A. Heilpern, Claire DiLeo, Rafael M. Almeida, Suresh A. Sethi, Marcela Miranda, ... , Joshua Fan, ... , Carla P. Gomes, & Alexander S. Flecker. ***Nature Sustainability***, 2025.

[4] Monitoring Vegetation from Space at Extremely Fine Resolutions via Coarsely-Supervised Smooth U-Net.

Joshua Fan, Di Chen, Jiaming Wen, Ying Sun, Carla Gomes.

IJCAI-22: *International Joint Conference on Artificial Intelligence (AI for Good track)*, 2022. **(Acceptance rate: 19.3%)**

- Also presented at the “Tackling Climate Change with Machine Learning” workshop at NeurIPS 2021

[5] A GNN-RNN Approach for Harnessing Geospatial and Temporal Information: Application to Crop Yield Prediction.

Joshua Fan*, Junwen Bai*, Zhiyun Li*, Ariel Ortiz-Bobea, Carla Gomes.

AAAI-22: *AAAI Conference on Artificial Intelligence (AI for Social Impact track)*, 2022. **(Acceptance rate: 15.0%)**

- Workshop version got **best paper award (ML Innovation)** at “Tackling Climate Change with ML” workshop, NeurIPS 2021.

[6] Scalable preprocessing for sparse scRNA-seq data exploiting prior knowledge.

Sumit Mukherjee, Yue Zhang, Joshua Fan, Georg Seelig, Sreeram Kannan. ***Bioinformatics***, 2018.

WORKSHOP PAPERS (peer-reviewed; underline denotes mentored students)

[7] A Novel Integrated ML Approach Utilizing Radar & Satellite Imagery for Selective Logging Detection.

Saraswathy Amjith, Joshua Fan. *Tackling Climate Change with ML workshop at NeurIPS 2025*.

[8] Detecting Aquaculture with Deep Learning in a Low-Data Setting.

L. Greenstreet, Joshua Fan, F. Pacheco, Y. Bai, M. Ummus, C. Doria, N. Barros, B. Forsberg, X. Xu, A. Flecker, C. Gomes. *Fragile Earth workshop at KDD 2023*. *Poster presented at AGU (American Geophysical Union) Annual Meeting, Dec 2023*.

PUBLICATIONS UNDER REVIEW (* denotes equal contribution; underline denotes mentored students)

[9] Locality and Distance-Aware Attention for Time Series Extrinsic Regression. *Submitted to AAAI 2026.*

Joshua Fan*, Kaitlyn Chen*, Shufeng Kong, Junwen Bai, Zhiyun Li, Ariel Ortiz-Bobea, Carla Gomes.

[10] Biogeochemistry-Informed Neural Network (BINN) for Improving Accuracy of Model Prediction and Scientific Understanding of Soil Organic Carbon. *Submitted to Geoscientific Model Development (GMD), 2025.*

Haodi Xu*, Joshua Fan*, Feng Tao*, Lifan Jiang, Fengqi You, Benjamin Z. Houlton, Ying Sun, Carla P. Gomes, Yiqi Luo.

[11] **AISciVision: A Framework for Specializing Large Multimodal Models in Scientific Image Classification.** *Arxiv*, 2024.

B. Hogan, A. Kabra, F. Pacheco, L. Greenstreet, **Joshua Fan**, A. Ferber, M. Ummus, A. Brito, ... , A. Flecker, C. Gomes.

PUBLICITY

Cornell AI for Sustainability Trainee Spotlight. Oct 2025.

AI-LEAF Institute spotlight interview. Oct 2025. High-level overview of my work in AI for agriculture and ecology.

POSTERS & TALKS

Scientifically-Interpretable Reasoning Network (SciReN) for Understanding the Soil Carbon Cycle.

Joshua Fan*, Haodi Xu*, Feng Tao, Md Nasim, Yiqi Luo, Carla Gomes.

Poster/talk at AI-LEAF Institute Annual Meeting, Fort Collins, CO, USA. May 2025.

Discovering Emergent Soil Carbon Mechanisms with a Biogeochemistry-Informed Neural Network (BINN).

Haodi Xu*, Feng Tao*, **Joshua Fan***, Yiqi Luo, Carla Gomes.

Poster/talk at AI-CLIMATE Institute Annual Meeting, Minneapolis, MN, USA. May 2024.

Detecting Aquaculture in the Brazilian Amazon using Deep Learning in a Low-Data Setting.

J. Fan, L. Greenstreet, F. Pacheco, Y. Bai, M. Ummus, C. Doria, N. Barros, B. Forsberg, X. Xu, A. Flecker, C. Gomes.

Poster at AGU (American Geophysical Union) Annual Meeting, San Francisco, CA, USA. Dec 2023.

Near Real-Time Crop Yield Forecasting with Interpretable Time-Series Deep Learning.

Talk at The Workshop on Environmental Economics and Data Science (TWEEDS), Eugene, OR, USA. Oct 2022.

Using Deep Learning to Monitor and Forecast Vegetation Growth.

Talk at Soil and Crop Sciences Seminar, Cornell University, Ithaca, NY, USA. Feb 2022.

Predicting Iron Bioavailability in Yellow Beans with Hyperspectral Imaging and Machine Learning.

Talk at Yellow Bean Conference, Online. Dec 2020.

TEACHING EXPERIENCE

- TA, Application of Machine Learning to Plant Science (Cornell University, PLSCI 7202): Fall 2022
- TA, Intro to Artificial Intelligence (Cornell University, CS 4700): Fall 2019
- TA, Probability & Statistics (UW, CSE 312): Fall 2015, Winter 2016, Spring 2017, Winter 2018, Winter 2019
- TA, Foundations of Computing I/Discrete Math (Univ of Washington, CSE 311): Fall 2016, Spring 2018
- TA, Intro to Machine Learning for Non-Majors (Univ of Washington, CSE 416): Spring 2019

MENTORSHIP

- **Kaitlyn Chen** (undergraduate, next position: Amazon) – Adding locality/distance inductive biases to Transformers for time-series regression.
- **Saraswathy Amjith** (high school, next position: MIT undergrad) – Detecting deforestation using deep learning with radar and optical satellite imagery.
- **Kevin Toepelt** (undergraduate, Luo lab) – domain adaptation to improve spatial generalization of BINN.

TECHNICAL SKILLS

- **Significant experience:** Python, Java, C#, SQL, C++, PHP/Hack
- **Working knowledge:** R, Matlab, HTML/CSS, JavaScript, Bash
- **Tools:** PyTorch, DGL, Pandas, Matplotlib, scikit-learn, rasterio, Git, Visual Studio, xarray, Google Earth Engine, ArcGIS

HONORS & AWARDS

- **NSF Research Training (NRT) Fellowship:** Digital Plant Science (2021-2024), AI for Sustainability (2025-2026)
- **Bob Bandes Memorial Excellence in Teaching Award**, University of Washington, 2019 (one of 3 winners)